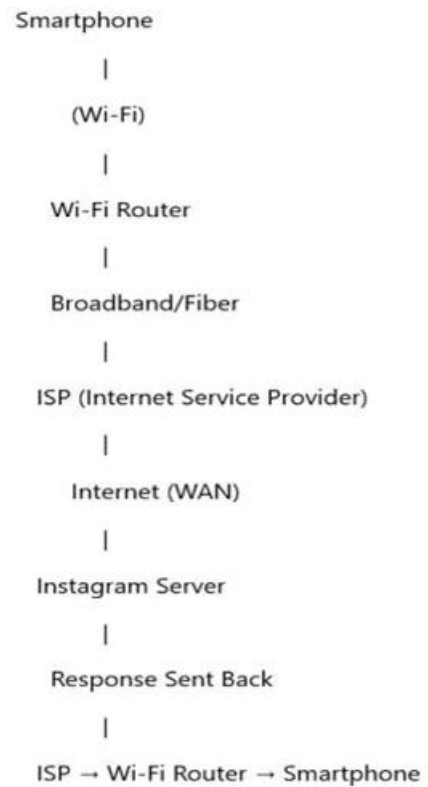


Introduction to networks

Task 1.

Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



Labels:

- Smartphone – User's device
- Wi-Fi Router – Connects devices to the internet
- ISP – Provides internet access
- Internet (WAN) – Global network connecting devices
- Instagram Server – Stores Instagram data and sends responses

Task 2.

List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Device	Network Type	Purpose
Smartphone	WLAN/WAN	Connects to Wi-Fi and mobile internet.
Laptop	LAN/WLAN	Connects to office/home Wi-Fi or Ethernet.
Wi-Fi Router	LAN/WAN	Connects local devices to the internet.
Smart TV	WLAN	Streams online videos through Wi-Fi.
Smartwatch	PAN	Connects to a smartphone using Bluetooth.

Task 3.

Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Internet:

The Internet is a worldwide public network that anyone can access. For example, when using public Wi-Fi at a railway station or café, you can browse websites like Google, YouTube, or Instagram.

Intranet:

An intranet is a private network used within an organization. For example, a college may have an internal portal where students can view attendance, assignments, or exam results. This portal is usually accessible only to students and staff with authorized login credentials.

Task 4.

Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

- When I open WhatsApp and search for a contact or message, the request starts from my smartphone.
- The phone sends the search request through Wi-Fi or mobile data to my Wi-Fi router or mobile network, then to my ISP and over the Internet to WhatsApp's servers (if needed).
- The server processes the request and sends the required data back through the Internet, ISP, and router to my smartphone.
- Finally, the app displays the search results on the screen within a few seconds.